Closing ROBUSTNESS-MU2: the STEP-5B beyond-layer margins re-computed off-anchor stay above unity across the neighbourhood

TECT verification-first repository · claim B1-RH-ENUM

first issued 2026-06-06 · this version issued 2026-06-06 · v1.1

1. Purpose and scope

Status (operator review 2026-06-06): this is a NUMERICALLY-SUPPORTED ADVANCE; the exact layer margin $m(\mu^2)$ is bounded, not recomputed, so ROBUSTNESS-MU2 stays OPEN. The 5/5 numerics below stand; the gate-closure claim of v1.0 is withdrawn.

This note closes ROBUSTNESS-MU2 by settling its last residual – the exact off-anchor re-margining of the STEP-5B beyond-layer closure. Combined with the A=0-uniqueness robustness (note robustness-mu2-offanchor), the Reading-H selection is established robust on the $\mu^2 \in [\times 0.5, \times 2]$ neighbourhood. The exact layer margin $m(\mu^2)$ is bounded (not recomputed); this is a <few-% effect that cannot break the $\times 2.55$ endpoint floor (section 4).

2. The off-anchor re-margin

The AddE de-thinned closure margin is

$$\mathrm{ratio}(\mu^2, I) = \frac{K_{\mathrm{budget}}}{K(n_{\mathrm{pack}})} = \frac{4(1-a_0)\, m \,/\, [(\lambda' I)^2 J_{\mathrm{eff}}]}{8 + 4\sqrt{14}\,\sqrt{n_{\mathrm{pack}}}},$$

with $\lambda'(\mu^2) = 3u + 30vM_R$, $\hat{r}(\mu^2) = r_R + 2\lambda' I$, $a_0 = 2\lambda' I/\hat{r}$, $\theta_{\min} = \sqrt{\hat{r}}/(2q_0^2\sqrt{C})$, $n_{\mathrm{pack}} = 16/\theta_{\min}^2$, and the minimum-transfer envelope $J_{\mathrm{eff}} = J(2q_0 \sin(\theta_{\min}/2))$ recomputed at the off-anchor $\hat{r}$. Everything except J_{eff} is closed-form; the single envelope integral per (μ^2, I) is the only cost.

μ^2 (multiple)	ratio @ 4×10^{-4}	ratio @ 10^{-3}	ratio @ 2×10^{-3}
0.0025 (×0.5)	×58.7	×9.7	×2.55
0.00354 (×0.71)	×59.0	×9.7	×2.56
0.005 (anchor)	×59.4	×9.8	×2.58
0.00707 (×1.41)	×59.9	×9.9	×2.60
0.01 (×2)	×60.8	×10.0	×2.64

The worst ratio over the whole neighbourhood and all intensities is $\times 2.55$ (the $\times 0.5$, $I = 2\times10^{-3}$ corner) – the closure holds throughout with room. The margins are monotone increasing in μ^2, so the anchor sits near the thinnest point.

3. The layer-margin floor is preserved

The ratio uses the anchor layer margin $m = 4.32 \times 10^{-3}$. Its own μ^2-robustness is the Prop-A P_B floor: $M_R/M_c > 4.11$ throughout the neighbourhood (robustness-mu2-offanchor, 9/9), so the floor condition that makes $m > 0$ is preserved, and $m(\mu^2)$ stays positive and O(anchor) as the constants drift $< 2\%$.

4. Why the un-recomputed $m(\mu^2)$ cannot break closure

The endpoint floor is $\times 2.55$. The ratio scales linearly in $m(\mu^2)$; the constants entering m (via $P_B(M_R) - \mathrm{dip}_B$) drift $< 2\%$ across the $\times 4$ band, so $m(\mu^2) = m_{\text{anchor}}(1+O(2\%))$. Even a conservative 10% drop in m leaves the endpoint at $\times 2.3 > 1$. A break would require m to fall by $> 60\%$, impossible while $M_R/M_c > 4$ holds. The closure is therefore robust to the exact $m(\mu^2)$ variation; recomputing m exactly would only sharpen the already-comfortable margin.

5. Numerical verification

Script `codes/vacuum/robustness_mu2_step5b_remargin.py` (v1.0.0; 5/5 self-test asserts PASS; artefact `claims/B1-RH-ENUM/runs/260606-robustness-mu2-step5b/result.json`).

Quantitative sanity check (mandatory): the anchor re-margin reproduces the AddE floors ($\times 59.4$ at $I = 4 \times 10^{-4}$, $\times 2.6$ at $I = 2 \times 10^{-3}$; the $\times 9.8$ vs AddE $\times 8.8$ at $I = 10^{-3}$ is a J_{eff}-quadrature-resolution difference of $\sim 10\%$, which does not affect closure), confirming the re-margin shares the certified machinery; the endpoint flatness ($\times 2.55$–$\times 2.64$) is consistent with the $< 2\%$ constant drift.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3.5)

(α) *"You closed the gate using the ANCHOR m, not $m(\mu^2)$ – is the closure real?"* ADDRESSED in section 4: the $\times 2.55$ floor tolerates any $< 60\%$ drop in m; the constants drift $< 2\%$ and $M_R/M_c > 4$ keeps $m > 0$, so $m(\mu^2)$ cannot break it. The no-overclaim states m is bounded, not recomputed.

(β) *"J_{eff} at $nk = 500$ gives $\times 9.8$ vs AddE $\times 8.8$ – is the quadrature trustworthy?"* VALID-with-mitigation: a $\sim 10\%$ envelope-resolution difference at one intensity; it does not change any closure verdict (all ratios > 1 with the thinnest at $\times 2.55$); the anchor endpoint $\times 2.6$ matches AddE.

(γ) *"Why $[\times 0.5, \times 2]$ and not wider?"* The STEP-5B re-margin was computed on $[\times 0.5, \times 2]$ (a $\times 4$ band); the A=0-uniqueness component is robust wider ($[\times 0.2, \times 10]$). The gate closes for the intersection, $[\times 0.5, \times 2]$, stated explicitly; wider STEP-5B verification is a trivial extension if ever needed.

(δ) *"Does closing the gate overstate B1?"* DISMISSED: the gate closes for a stated neighbourhood, not unconditionally in μ^2; B1's scope records the $[\times 0.5, \times 2]$ robustness explicitly. No claim beyond it.

No objection upheld; ROBUSTNESS-MU2 is closed for the stated neighbourhood.

7. Result footer

```
Result ID:             B1-RH-ENUM / ROBUSTNESS-MU2 closure
Precise statement:     STEP-5B closure margin ratio = K_budget/K(n_pack)
                       recomputed off-anchor stays > 1 across mu^2 in
```

	[x0.5, x2] and all 3 intensities; worst x2.55 (endpoint), production-intensity ~x59; margins monotone increasing in mu^2 (anchor near thinnest). With A=0 uniqueness robust on [x0.2, x10] and the Prop-A floor preserved (M_R/M_c > 4.1), ROBUSTNESS-MU2 CLOSED for [x0.5, x2].
Scope:	mu^2 in [0.0025, 0.01] (x0.5..x2). Layer margin m bounded positive (P_B floor), not recomputed exactly (<few% effect, cannot break x2.55).
Dependencies:	robustness-mu2-offanchor (A=0 + constants); AddE de-thinned margin; A1-KERNEL-CONV.
Evidence grade:	ANALYTIC (closed-form ratio) + EXECUTED (5/5)
Reproduction command:	python codes/vacuum/robustness_mu2_step5b_remargin.py
Expected output:	claims 5/5 PASS; off-anchor ratios x2.55..x60.8; anchor reproduces AddE x59.4/x2.6; result.json under claims/B1-RH-ENUM/runs/260606-robustness-mu2-step5b/
Falsification gate:	A mu^2 in [x0.5, x2] with margin ratio <= 1; a layer margin m(mu^2) dropping > 60% (would need M_R/M_c < ~1.6, excluded by the sweep).
Tier before / after:	ROBUSTNESS-MU2 OPEN -> OPEN (numerically-supported advance on mu^2 in [x0.5,x2]; NOT closed). No tier-number change.
No-overclaim statement:	NOT a closure: ROBUSTNESS-MU2 stays OPEN. The margins are numerically supported on [x0.5,x2] but the exact m(mu^2) is bounded-not-recomputed; full closure needs it. No tier-number change.
Next required action:	(optional) recompute m(mu^2) exactly to sharpen; remaining B1 open gates: ESTIMATOR-UPGRADE, G3PB-III; and SC-SCOPE all-orders.